Heaven Alakraa

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EDUCATION

Brown University, *Sc.B in Mechanical Engineering, GPA: 3.96/4.00* **Providence, RI | Aug 2024 - May 2028 (Expected)**

Relevant Courses: *Engineering Statics and Dynamics; Mechanics of Solids and Structures; Computer Aided Visualization and Design; Biomaterials; Thermodynamics; Fluid Mechanics; Electrical Circuits and Signals; Control Systems Engineering; Electricity and Magnetism; Biomechanics.*

Honors: 2026 Carlo De Luca Biomechanics Summer Undergraduate Research Experience (B-SURE) Award, American Society of Biomechanics.

PROFESSIONAL EXPERIENCE

Engineering Research Intern | *Bioengineering Laboratory, Dept. of Orthopaedics,* **Providence, RI May 2026 – Present**

- Design passive and active pendulum testing apparatus in SolidWorks to measure the coefficient of friction in porcine wrist joint specimens via swing-decay analysis, built full 3D model, 2D drawing package, and detailed bill of materials
- Fabricate and assemble the experimental testing device at the RIH Orthopaedic Foundation testing facility, conduct mechanical validation to characterize tribological properties of soft tissue structures under physiological loading conditions

Product Design Intern | *Lilac Biosciences Inc.* **Providence, RI | Jan 2026 – Present**

- Redesign RNA purification tube geometry in SolidWorks to replace centrifuge-dependent spin-column separation with filtration integrated directly into the tube, developing the internal flow path and membrane seat from concept through CAD
- Evaluate filtration membrane candidates through structured literature review and vendor analysis to identify optimal pore size and material compatibility for RNA retention, fabricate and iterate prototypes via 3D printing for experimental validation

Cooling and Radiator Testing Engineer | *Brown Formula Racing Team (Brown FSAE),* **Providence, RI | Sep 2025 – Present**

- Designed a wind-tunnel test rig in SolidWorks for radiator thermal-performance characterization and sourced piping and fluid-system components via McMaster-Carr to support fabrication of the cooling system testing apparatus
- Measured radiator heat-transfer coefficients through 3-bucket calorimetry and mass-flow-rate testing, generating experimental data to quantify heat transfer coefficients and inform cooling-system redesign.

Biomedical Research Assistant | *Tripathi Lab: Biomedical Engineering at Brown University,* **Providence, RI | Jan 2025 – Sep 2025**

- Quantified AAV capsid and transgene concentrations using NanoMosaic's Tessie nanoimaging platform, co-author in pending publication: "Overcoming an Analytical Bottleneck in Gene Therapy: A Rapid and Precise Nanoneedle ImmunoAssay Method for Detection and Quantification of Adeno-Associated Virus Capsid and Transgene Titers in One System"
- Performed DNA extraction, bisulfite conversion, BioAnalyzer QC and DNA methylation analysis for epigenetics research.

PROJECTS

Can Opener — Mechanical Design, Documentation & Analysis | *SolidWorks, FEA, Motion Analysis, ASME Y14.5*

- Modeled a fully constrained 10-component rotary can-opener assembly in SolidWorks and produced a 6-sheet 2D drawing package (ASME Y14.5-2009, Critical-to-Function dimensioning) with section, detail, and exploded views
- Ran static FEA under a 200 N grip load in AISI 304 stainless steel, confirming stresses remained well below the 207 MPa yield strength and identifying a stress-concentration artifact at an idealized boundary condition
- Built a motion study of the knob-driven gear train to verify continuous gear mesh and quantify mechanical advantage

Cantilever Beam Deflection Scale | *MATLAB, Fabrication, Brown Design Workshop*

- Designed and fabricated a cantilever beam-based mechanical scale, applying Euler–Bernoulli beam theory and Hooke's Law to translate analytical deflection expressions into a physical model using bandsaw and wood drill tools in the BDW.
- Optimized beam dimensions and load placement using MATLAB, computing spring constant, bending stress, and safety factor, validated design through experimental incremental calibration, and achieved <10% error predicting unknown mass

Next-Gen Material Selection — Vertebral Body Replacement | *Ansys Granta EduPack, FEA*

- Translated clinical requirements and FEA outputs into quantitative mechanical and biological screening criteria for implant material selection, screened 4,395 candidate materials in Ansys Granta EduPack and identified polyetherimide (PEI) as a leading candidate for future clinical vertebral body replacement applications.

Heart Rate Sensor & Formula 1 Safety Gear Testing | *Arduino, MATLAB, Prototyping*

- Prototyped a real-time heart-rate monitor using MAX30102 pulse oximetry and an Arduino OLED display, analyzed and visualized BPM data in MATLAB across simulated F1 safety-gear conditions, and presented findings and live demonstration

LEADERSHIP & COMMUNITY DEVELOPMENT

Teaching Fellow | *Brown University - National Education Opportunity Network* **Providence, RI | Aug 2025 - Present**

- Deliver instruction and design mentorship to 200+ students in *Introduction to Engineering: Design, Materials, and Structures*

Operations and Administrative Coordinator | *Student Accessibility Services* **Providence, RI | Mar 2025 – Present**

- Lead front-line support for 7000+ students, improving process efficiency and service delivery under tight academic deadlines.

TECHNICAL SKILLS

CAD & Design: SolidWorks, Fusion 360, Onshape, 2D Engineering Drawings, GD&T ASME Y14.5, Design for Manufacturing
Simulation & Analysis: FEA, Abaqus, Motion Analysis, CAM Analysis, Computational Fluid Dynamics (CFD), LTspice, Simulink
Programming & Data Analysis: MATLAB, Python, Arduino, ImageJ, PyCharm, Pandas